Path Finding for Mobile Robots Directly from Natural Language Instructions

An Undergraduate Project Proposal

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Abstract

Classical path planning assumes that a robot's task can be reduced to a single goal coordinate. Human instructions are considerably richer: they specify regions to avoid, intermediate places to visit, orderings between them, and soft preferences that a planner is expected to honour where convenient but not at any cost. The prevailing approach to bridging this gap translates the instruction into a formal temporal logic specification before planning. That route is well studied, but it is brittle to translation error, opaque to non-expert users, and structurally unable to represent graded preferences. The opposite extreme, asking a large language model to emit the trajectory itself, discards the correctness guarantees that search algorithms provide and produces physically invalid paths at a high rate. This proposal identifies the intermediate regime as an underexplored problem: interpreting free-form navigation instructions without constructing a formal specification, while retaining the validity and optimality guarantees of classical search. We set out the problem, argue why neither existing extreme resolves it, and outline at a high level the direction we intend to investigate.

1. Background and Motivation

Path planning is among the most thoroughly studied problems in robotics. Given an occupancy map, a start state and a goal state, algorithms such as A* and its numerous descendants return a collision-free path and, under an admissible heuristic, guarantee that the path is optimal with respect to the stated cost function. These guarantees are the reason such methods remain the backbone of deployed navigation stacks decades after their introduction.

The guarantees, however, are purchased at the price of an extremely narrow task interface. A planner accepts a goal coordinate and nothing else. Every other aspect of what the operator wanted must be pre-encoded into the cost map by an engineer before deployment. As mobile robots move out of controlled industrial settings and into hospitals, warehouses, campuses and homes, the people issuing instructions are increasingly not engineers, and the instructions they issue are not coordinates.

Consider an instruction of a kind that any building occupant might give:

> *"Go to the lab, but do not cut through the workshop, and stop at the store room on the way."*

A single sentence of this form carries at least four distinct pieces of information: a destination, a hard prohibition over a named region, an intermediate subgoal, and an implied temporal ordering between the subgoal and the destination. A conventional planner can accept the first and is structurally incapable of accepting the remaining three. Closing this gap, without surrendering the guarantees that make classical planning trustworthy, is the motivation for this project.

2. Problem Statement

We state the problem as follows.

Given a two-dimensional map of a static environment in which traversable and blocked cells are known and in which certain connected regions carry human-meaningful labels, together with a free-form natural language instruction issued by a non-expert user, **produce** a path from the robot's current position that (i) is guaranteed to be physically traversable, (ii) satisfies every hard constraint stated in the instruction, (iii) respects soft preferences stated in the instruction to the extent that doing so is not disproportionately costly, and (iv) is obtained without

constructing an intermediate formal logic specification.

Clause (iv) is a deliberate restriction rather than an incidental one, and requires justification. It is given in Section 3.

2.1 Characterising the instruction space

Not all navigation instructions are equally difficult, and the difficulty is not primarily linguistic. We distinguish the following classes, in increasing order of the demand they place on a planner.

Class	Representative form	Planner demand
Destination	*Go to the lab.*	Goal specification only.
Prohibition	*… without entering the workshop.*	Hard region exclusion.
Subgoal	*Stop at the store room on the way.*	Intermediate visit obligation.
Ordering	*Visit A before B.*	Temporal sequencing of subgoals.
Soft preference	*Stay in the corridor if you can.*	Graded, violable cost bias.

Table 1: Classes of navigation instruction considered in this project.

The final class deserves particular attention. A soft preference is not a weak constraint that may be dropped when infeasible; it is a constraint whose satisfaction is a matter of degree and whose enforcement must be traded against path cost. This distinction turns out to be central to the argument of Section 3.

3. Why the Existing Approaches Are Unsatisfactory

3.1 The formal specification route

The dominant line of work translates the natural language instruction into a temporal logic formula, most commonly Linear Temporal Logic, and then invokes a planner capable of synthesising a trajectory satisfying that formula. This is a mature approach with an appealing property: once the formula is obtained, satisfaction can be verified formally.

Three objections nonetheless apply. First, the pipeline is brittle in a particular way — a translation that is almost right is not partly useful but simply wrong, and the resulting plan may be confidently executed while bearing no relation to what the user asked. Second, the intermediate artefact is unreadable to the very users the natural language interface exists to serve, so a user cannot inspect the system's understanding of their own instruction before execution. Third, and most fundamentally, temporal logic is a Boolean formalism: a formula is either satisfied or violated. Instructions in the soft preference class of Table 1 have no faithful Boolean encoding. They can only be admitted by discretising a graded notion into an arbitrary threshold, which changes the instruction rather than representing it.

3.2 The end-to-end language model route

The opposing approach dispenses with formal machinery and presents the map and instruction directly to a large language model, requesting the path as output. This is attractive in its simplicity and handles linguistic variation gracefully. Its failure is equally clear. Language models perform poorly at fine-grained spatial reasoning over coordinate grids; reported behaviour includes paths that traverse obstacles, contain discontinuities, or terminate short of the goal. More seriously, the approach abandons the guarantees that motivate the use of search algorithms in the first place. A path that may or may not be traversable is not a weaker result than a guaranteed one; for a physical robot it is a different and unusable kind of object.

3.3 The gap

The two approaches fail in complementary directions. The first is rigorous but linguistically and expressively rigid; the second is linguistically flexible but offers no correctness guarantee whatsoever. The problem this project addresses is whether the intermediate regime can be occupied: whether the interpretation of the instruction can be separated cleanly from the construction of the path, such that flexibility is obtained at the interpretation stage and guarantees are retained at the construction stage. To our knowledge this separation has not been systematically characterised or evaluated for the instruction classes of Table 1.

4. Proposed Direction

We outline the intended direction at a high level; the detailed design is deliberately left open at this stage and will be settled during the initial phase of the project.

Our working hypothesis is that the two capabilities being conflated by both existing approaches — understanding what a sentence asks for, and computing a route that provides it — are distinct and should be assigned to distinct components. Language understanding is the task at which language models are strong and search algorithms are absent; spatial route computation is the task at which search algorithms are reliable and language models are weak. We therefore intend to investigate architectures in which a language component is responsible solely for extracting the user's intent into a compact, human-inspectable intermediate representation, and a classical search component is responsible solely for producing the path under that representation. Critically, the language component would at no point emit a coordinate belonging to the final path.

Should such a separation hold, an immediate structural consequence follows: errors in language understanding could render a path incorrect with respect to the instruction, but could not render it physically invalid, because validity is determined entirely by the search component. Establishing whether this consequence is realised in practice, and quantifying the residual error, is the principal empirical question of the project.

We intend to constrain the intermediate representation to remain non-symbolic in the temporal logic sense, in keeping with clause (iv) of Section 2, and to be legible to a non-expert user so that the system's interpretation can be inspected before execution.

5. Evaluation Plan

No standard benchmark exists for the instruction classes of Table 1 with checkable ground truth, so constructing one is itself part of the work. We propose to evaluate along the following axes.

- **Path validity.** The proportion of returned paths that are in fact traversable and terminate at the intended destination. This isolates the guarantee that the end-to-end approach forfeits.
- **Instruction satisfaction.** The proportion of paths meeting every hard constraint stated in the instruction, assessed by automatic checkers written against each instruction rather than by inspection.
- **Cost of compliance.** The ratio of the returned path's cost to that of the cheapest path satisfying the same instruction, quantifying whether obedience is being bought with unreasonable detours.
- **Robustness to paraphrase.** Whether semantically equivalent rewordings of one instruction yield equivalent behaviour, as a direct measure of brittleness.
- **Failure attribution.** For each failure, the stage responsible. A modular architecture should permit this attribution; a monolithic one cannot, and the resulting diagnostic breakdown is regarded as a result in its own right.

Comparison will be made against two reference points: a planner that ignores the instruction entirely, establishing the floor below which no method may be considered to have contributed anything; and an end-to-end language model asked to emit the path directly, representing the unstructured extreme.

6. Scope and Delimitations

We state the boundaries of the work explicitly, so that its claims are not read more broadly than intended.

- Evaluation is conducted in simulation on two-dimensional grid environments. No physical robot, sensing or control is involved.
- A single robot is considered. Multi-robot coordination, and the collision constraints it introduces, are outside the scope.
- Environments are static. Dynamic obstacles and online replanning are not treated.
- Semantic region labels are assumed available. Perception, mapping and semantic segmentation are treated as solved inputs.
- Instructions are assumed well formed and unambiguous. Detecting ambiguity and issuing clarification requests is a substantial problem in its own right and is not attempted here.

7. Anticipated Outcomes

- A characterisation of the natural language path planning problem for instruction classes that formal specification methods represent poorly, particularly graded preferences.
- A reusable evaluation set of labelled environments and instructions paired with automatic satisfaction checkers.
- Empirical evidence on whether separating instruction interpretation from path construction preserves classical planning guarantees, and quantification of the error that remains.
- A diagnostic account of the failure modes encountered, attributed by stage.

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